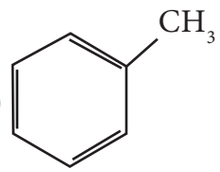
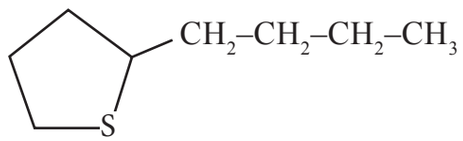
Based on the above classification let us classify the following compounds.

1. Classify the following compounds based on the structure

i) $CH \equiv C- CH_2-C \equiv CH$

ii) $CH_3 - CH_2 - CH_2 - CH_2- CH_3$

iii) 

iv) 

Solutions:

(i) Unsaturated open chain compound

(ii) saturated open chain compound

(iii) aromatic benzenoid compound

(iv) alicyclic compound

Evaluate Yourself

1) Give two examples for each of the following type of organic compounds.

(i) non-benzonoid aromatic,

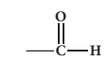
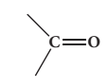
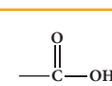
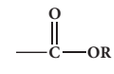
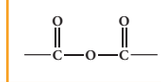
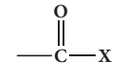
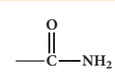
(ii) aromatic heterocyclic,

(iii) alicyclic and

(iv) aliphatic open chain.

11.2.2 Classification based on functional groups:

Table 11.1 Class of compounds and their functional group

S. No.	Class of compounds	Functional group	General formula (R - Alkyl group)
1	Alkene	–	C_nH_{2n}
2	Alkyne	–	C_nH_{2n-2}
3	Alkyl halide	—F, —Cl, —Br, —I	R- X
4	Alcohol	—OH	R-OH
5	Ether	—O—	R-O-R'
6	Aldehyde		R-CHO
7	Ketone		R-CO-R'
8	Carboxylic acid		R-COOH
9	Ester		RCOOR'
10	Acid anhydride		R-CO-O-CO-R'
11	Acyl halide		R-COX
12	Sulphonic acid	—SO_3H	R-SO_3H
13	Nitro alkane	—NO_2	R-NO_2
14	Amine	—NH_2	R-NH_2
15	Amide		R-CO-NH_2

Unit 11 revised - jagan.indd 113 22-01-2021 12:42:14